

ARAVIND CHOWDARY KOLLI

QA Automation Lead | Test Automation Architect | DevOps Quality Engineer

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PROFESSIONAL SUMMARY

- QA Automation Leader with 11+ years of IT experience and 8+ years specializing in automation architecture, quality engineering, and DevOps-driven testing strategies within Banking and Financial Services domains.
- Proven expertise in designing scalable enterprise automation frameworks across UI, API, microservices, and cloud-native systems.
- Strong background in CI/CD-integrated automation, performance validation, database reconciliation, and distributed systems testing.
- Experienced in building Automation Centers of Excellence, mentoring QA teams, improving automation coverage to 85%+, and reducing production defect leakage by over 40%.
- Deep expertise in AWS cloud testing environments, containerized automation execution, and shift-left testing strategies aligned with Agile delivery models.
- Extensive experience leading end-to-end test strategy planning across multiple enterprise programs.
- Strong hands-on expertise in Selenium, Cypress, Playwright, Rest Assured, and BDD-based automation frameworks.
- Proven ability to implement automation governance models and quality metrics dashboards for executive visibility.
- Skilled in validating event-driven architectures and asynchronous microservices communication.
- Experience implementing parallel test execution strategies reducing regression cycle time by over 55%.

- Strong expertise in performance and load testing using JMeter for high-transaction financial systems.
- Adept at integrating automation frameworks with Jenkins, GitHub Actions, and GitLab CI pipelines.
- Solid understanding of application security validation, including OAuth 2.0, JWT authentication, and OWASP best practices.
- Experienced in backend data validation, reconciliation testing, and complex SQL-based verification.
- Proven ability to collaborate with architects, developers, and DevOps teams to embed quality into early design phases.
- Strong analytical and troubleshooting skills with expertise in root cause analysis and defect prevention strategies.

TECHNICAL SKILLS

Programming Languages : Java, JavaScript, TypeScript, SQL, Bash

UI Automation : Selenium WebDriver, Cypress, Playwright, Hybrid Framework, Page Object Model (POM), Data-Driven Framework

API Automation : Rest Assured, Karate, Cucumber (BDD), Postman, Contract Testing, Swagger Validation

Performance Testing : Apache JMeter, Load Testing, Stress Testing, Throughput Analysis, SLA Validation

CI/CD & DevOps : Jenkins, GitHub Actions, GitLab CI, Docker, Kubernetes, Maven, Gradle, SonarQube, Git

Cloud Platforms : AWS (EC2, S3, Lambda, CloudWatch), AWS-based Test Environments

Databases : SQL Server, Oracle, DB2, MongoDB, NoSQL, Database Reconciliation & Backend Validation

Monitoring & Reporting : Allure Reports, Prometheus, Grafana, ELK Stack

Security Testing : OWASP Validation, OAuth 2.0, JWT Authentication Testing, Role-Based Access Validation

Testing Practices : TDD, BDD, Shift-Left Testing, Agile Testing, Smoke Testing, Regression Testing, Integration Testing, E2E Testing

Test Management Tools : JIRA, Azure DevOps Boards, TestRail

PROFESSIONAL EXPERIENCE

Client: Morgan Stanley, New York, NY

Role: QA Automation Lead

Duration: Mar 2021 – Present

Led enterprise-wide QA automation strategy across distributed financial microservices architecture.

Key Contributions:

- Architected unified hybrid automation framework supporting UI, API, and DB validation layers.
 - Directed automation strategy across 15+ microservices in cloud-native environment.
 - Implemented Selenium Grid & parallel execution reducing regression runtime by 60%.
 - Built API automation suite integrated with Jenkins CI pipelines.
 - Developed BDD automation framework using Cucumber and Gherkin.
 - Designed advanced database reconciliation scripts validating transactional integrity.
 - Containerized automation execution using Docker for CI consistency.
 - Integrated automation reporting dashboards using Allure & Jenkins plugins.
 - Introduced automation governance and code quality standards.
 - Implemented performance validation tests using JMeter for peak traffic simulation.
 - Led 7-member QA automation team and mentored junior engineers.
 - Increased automation coverage from 50% to 85%.
 - Reduced production defect leakage by 40%.
 - Implemented shift-left testing strategy within sprint lifecycle.
 - Automated security validation for authentication and authorization workflows.
 - Conducted root cause analysis and introduced preventive automation controls.
 - Established quarterly automation framework health audits.
 - Provided executive-level reporting on automation ROI and quality metrics.
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- Defined enterprise-wide automation roadmap aligned with DevOps transformation and cloud modernization initiatives.
 - Collaborated with solution architects and product owners to embed quality gates during system

design phase.

- Implemented service virtualization strategies to enable early API testing before dependent systems were available.
- Established automated risk-based regression strategy prioritizing high-impact financial workflows.
- Designed reusable automation accelerators to support onboarding of new microservices.
- Introduced automation analytics dashboards tracking flaky test trends and stability metrics.
- Implemented environment provisioning validation to ensure readiness before regression execution.
- Led cross-functional defect triage meetings to improve issue resolution turnaround time.
- Standardized automation branching and versioning strategies aligned with Git workflows.
- Spearheaded automation framework scalability enhancements to support multi-release parallel testing cycles.

Environment: Selenium, Cypress, Java, Rest Assured, AWS, Jenkins, SQL Server, Oracle, Docker, Kubernetes

Client: Capital One, Richmond, VA

Role: Senior QA Automation Engineer

Duration: Jan 2018 – Feb 2021

Delivered automation solutions for online banking systems and microservices-based transaction platforms.

Key Contributions:

- Designed Selenium + Java automation framework using POM architecture.
- Built comprehensive API automation suite for Spring Boot microservices.
- Developed reusable automation utilities and test data management modules.
- Automated regression suite reducing manual testing effort by 70%.
- Performed backend database validation for DB2 and MongoDB systems.
- Conducted performance testing using JMeter for transaction-heavy applications.
- Integrated automation pipelines with GitLab CI & Jenkins.
- Validated OAuth 2.0 and JWT authentication flows.
- Implemented cross-browser and cross-environment validation strategy.
- Migrated 400+ manual test cases into automated scripts.
- Implemented test data masking and anonymization utilities.
- Introduced headless browser testing for faster CI execution.
- Enhanced automation stability using retry mechanisms & error categorization.
- Supported AWS cloud migration testing initiatives.
- Conducted knowledge-sharing sessions across QA teams.

- Designed scalable automation framework architecture supporting modular test case development and easier maintenance.
- Implemented data-driven and keyword-driven testing approaches to improve automation flexibility.
- Developed API contract testing strategy to validate backward compatibility across microservices.
- Built reusable API validation libraries for standardized request/response verification.
- Integrated automation reports with Allure and custom dashboards for enhanced visibility.
- Automated negative, boundary, and edge-case scenarios to improve defect detection in early stages.
- Collaborated with DevOps teams to enable automated smoke testing for every build deployment.
- Optimized regression execution time through test suite categorization and parallel execution strategy.
- Implemented environment-specific configuration management for multi-environment testing (DEV, QA, UAT, PROD).
- Conducted root cause analysis for flaky tests and introduced stabilization mechanisms.
- Assisted in defining QA KPIs including automation coverage, pass rate trends, and defect leakage metrics.
- Participated in architectural discussions to improve system testability and enhance automation readiness.

Environment: Selenium, Java, TestNG, Rest Assured, JMeter, GitLab CI, DB2, MongoDB, AWS

Client: TransUnion, Chicago, IL

Role: QA Automation Engineer

Duration: Jun 2014 – Dec 2017

Supported credit reporting systems with UI and API automation initiatives.

Key Contributions:

- Developed UI automation scripts for Angular-based credit monitoring platform.
- Built REST API validation suites using JUnit and Rest Assured.
- Integrated automation with Jenkins for nightly regression execution.
- Performed backend SQL validation for large-scale credit data.
- Designed cross-browser compatibility testing strategy.
- Automated regulatory compliance validation scenarios.
- Participated in integration testing across distributed systems.
- Stabilized regression suite by improving error handling & logging.
- Assisted in CI pipeline setup for automated regression testing.
- Supported UAT cycles and resolved production defects.
- Implemented structured logging and screenshot capture for better diagnostics.
- Contributed to early-stage automation framework enhancements script duplication across projects.
- Implemented data-driven testing approach to validate multiple credit scenarios using external test data sources (Excel/CSV/JSON).
- Created dynamic XPath and CSS locator strategies to improve script stability across UI updates.
- Automated negative and boundary test scenarios to improve defect detection in early testing phases.
- Validated batch data processing jobs by performing backend database reconciliation checks.
- Enhanced Jenkins pipeline by configuring automated email notifications and failure alerts for regression runs.
- Collaborated closely with developers to resolve flaky test cases and improve application testability.
- Participated in defect root cause analysis sessions to identify recurring functional gaps and improve automation coverage.
- Improved automation execution time by optimizing test suite grouping and selective execution strategies.
- Maintained detailed automation documentation and traceability mapping between test cases and business requirements.

Environment: Selenium, Java, JUnit, Jenkins, SQL Server, MongoDB

KEY AUTOMATION INITIATIVES

- Migrated organization from 30% to 85% automation coverage.
 - Reduced release cycles by 35% through CI-driven automation integration.
 - Implemented parallel execution reducing regression runtime by 55%.
 - Designed reusable hybrid automation framework (UI + API + DB).
 - Standardized QA automation governance across multiple teams.
 - Improved defect detection rate before production by 45%.
 - Built centralized automation reporting dashboard for leadership visibility.
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- Designed centralized test utilities for logging, screenshot capture, and failure diagnostics to improve debugging efficiency.
 - Implemented CI pipeline quality gates to prevent builds from progressing when critical test failures occur.
 - Developed automated database reconciliation scripts to validate transactional integrity across microservices.
 - Introduced tagging and selective test execution strategies to support faster release cycles.
 - Conducted automation framework refactoring to reduce technical debt and improve maintainability.
 - Integrated containerized test execution using Docker for consistent and isolated test environments.
 - Implemented API response time validation checks to ensure performance SLAs were consistently met.
 - Established automation onboarding documentation and coding standards for new QA engineers.
 - Designed automated health-check validation for dependent services before executing integration tests.
 - Contributed to shift-left testing initiatives by supporting early-stage API validation during development cycles.

EDUCATION

Master of Science – Software Engineering
2017 – 2019

Bachelor of Science – Information Technology
2012 – 2016

CERTIFICATIONS

- ISTQB Advanced Level Test Analyst
- Certified Selenium Automation Engineer
- AWS Certified Cloud Practitioner
- Certified Scrum Master (CSM)